Session Objectives

At the conclusion of the session/course, the student should be able to:

- Describe the presentation and diagnostic criteria used to diagnose Systemic Lupus Erythematosus (SLE).
- Describe the different uses of ANA testing.
- Describe the presentation of Systemic Sclerosis (Scleroderma)
- Describe the presentation of Sjorgen Syndrome

Source: Course Syllabus

Autoimmunity Considerations

- Considerations:
  - Chronic, relapses, remissions, progressive, systemic…
- Nature of underlying immune Response
- Hypersensitivity Reactions:
  - II: antibodies bind to cell surface causing destruction
  - III: Immune complex- antigen- antibody -complement

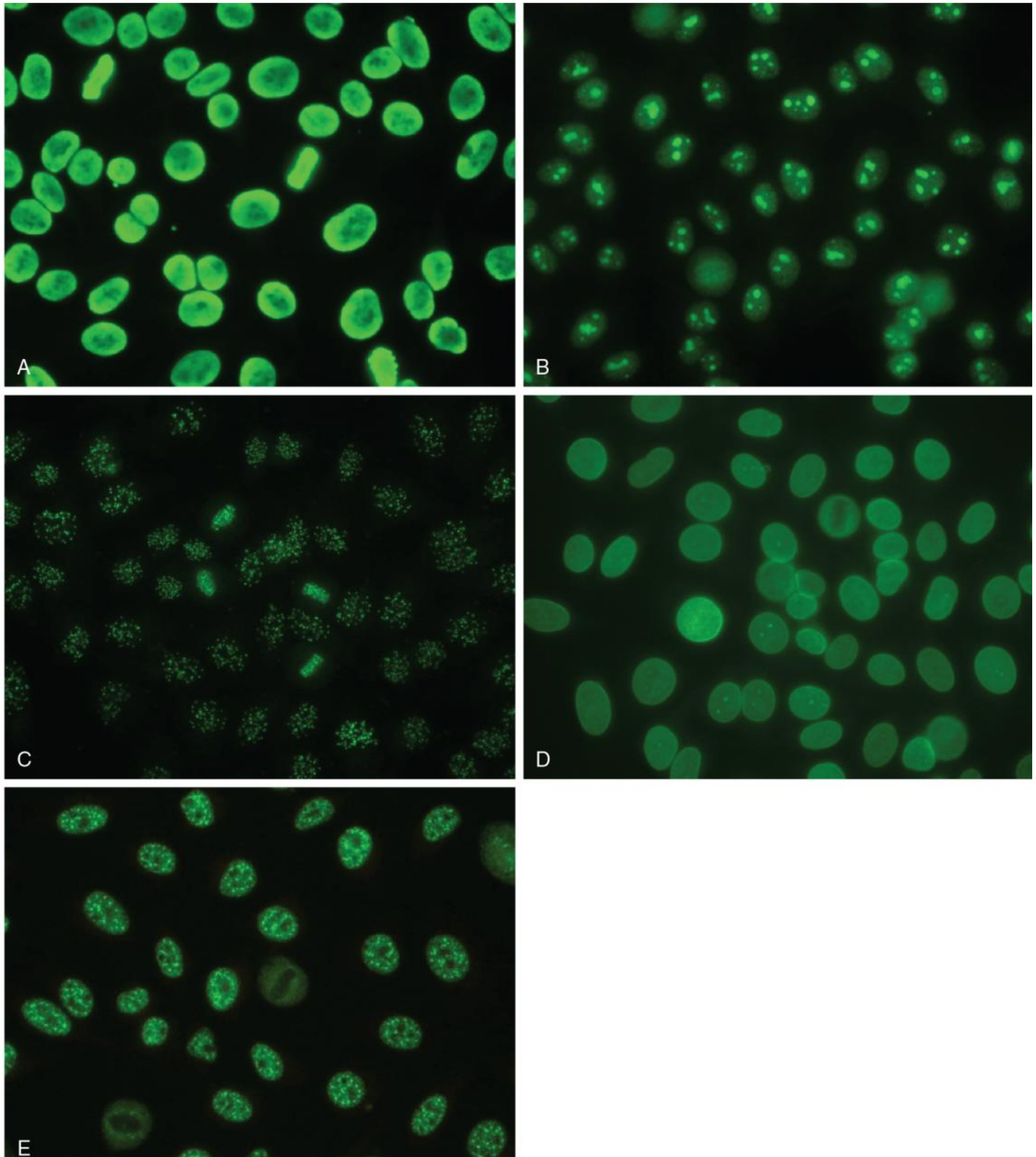
Systemic Lupus Erythematosus (SLE)

- Multiple organ/system disorder
  - Effects mostly from Ab-Ag complex **deposition** in the capillaries of organs *(Hypersensitivity type III)*
- Autoimmune, inflammatory disorder
  - Auto-Ab to nuclear antigens
- Remission-relapse cycle
- May be mild and self-limited or can progress to life threatening end organ damage

SLE Pathogenesis

- Genetic Predisposition
  - B and T cells specific for self nuclear antigens
    - Defective tolerance
  - Defective clearance of apoptotic bodies
- Environmental Factors:
  - UV light
  - Drugs: hydralazine, procainamide, D-penicillamine.
- Harm: immune complex deposition and binding of antibodies..

Staining Patterns of Antinuclear antibodies



(A) Homogeneous staining of nuclei is typical of antibodies reactive with double-stranded DNA (dsDNA) and nucleosomes and is common in systemic lupus erythematosus (SLE). (B) Nucleolar clumpy pattern is typical of antibodies against the nucleolar protein fibrillarin (U3-RNP) and is strongly associated with systemic sclerosis. (C) Centromere pattern is seen predominantly in systemic sclerosis, primary biliary cirrhosis, and Sjögren syndrome. (D) Nuclear envelope pattern is caused by antibodies against nuclear membrane proteins, such as gp210, p68, and lamins, and can occur in patients with autoimmune liver diseases. (E) Coarse speckled pattern is seen with antibodies against Sm and U1-RNP, and can be observed in SLE, systemic sclerosis, and mixed connective tissue disease.

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Non-pharmacologic Treatment Planning for SLE

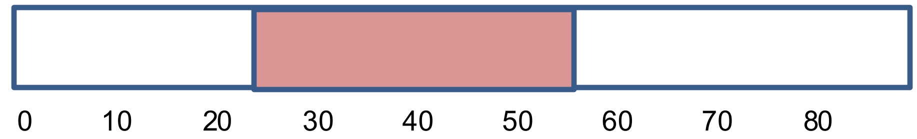
- Counselling and emotional support
- Multidisciplinary team approach

- Avoidance of UV light
- Generalized health promotion
- Screening for cardiovascular and renal disease
- Exercise and weight control
  - Osteoporosis
  - Medication side effects

Systemic Sclerosis: Epidemiology

Sex:

3:1 F:M



Onset at 25-55 years

Peak 50-60

Who

Race/Ethnicity:
Earlier and more severe presentation in African-American patients

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What is Systemic Sclerosis (Scleroderma)?

- Fibrosis of skin and internal organs
- Raynaud phenomenon is present in almost all patients with scleroderma
- Skin effects can be limited or diffuse
- Systemic effects: GI, Pulmonary, heart, muscles, and Renal

Treatment Plans of Systemic Sclerosis

Systems affected

- Raynauds
  - M/C treatment – Calcium channel blocker
- Esophageal Motility
  - Acid reduction (PPIs)
  - Oral prokinetic agents
    - Erythromycin
    - Metoclopramide
- Sclerodactyly
  - OT, surgical options

Diffuse treatments

- Methotrexate
- Cyclophosphamide
- IVIG
- Rituximab
- Tocilizumab

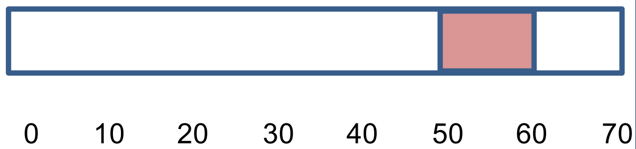
- Prednisone does NOT have a role in treatment: increases risk of renal crisis

Overview of treatment of Systemic Sclerosis

- Treatment is typically based on symptoms and organ systems effected
- Counselling needed for support throughout after diagnosis
- Nutritional support and guidance needed
- Discussion of prognosis encouraged to assist patients with understanding of disease progression.

Key points for Sjogren's Syndrome

Sex:
16:1 F:M



Most common in females between age of 50-60.
50% have another autoimmune disorder

Diagnostic Criteria:

- Salivary gland inflammation by biopsy
- **Anti Ro/SSA or Anti La/SSB Abs**
- Oral signs or ocular signs of dry mouth or eyes for 3 months

<u>Pathogenesis:</u>

Gland: inflammatory infiltrates. Mix of polyclonal T and B cells early in course.
Late in course: dominant B-Cell Clone.
5% develop lymphoma

Key targets

- Exocrine glands with decreased secretory function
  - Lacrimal glands
  - Salivary glands
- Can also effect lungs, thyroid, kidneys, skin and muscle

Source: Brandt JE, Priori R, Valesini G, Fairweather D. Sex differences in Sjögren's syndrome: a comprehensive review of immune mechanisms. *Biol Sex Differ*. 2015;6:19. Published 2015 Nov 3. doi:10.1186/s13293-015-0037-7

Key points for Sjogren's Syndrome

Key targets

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Key Symptoms:

- Keratoconjunctivitis sicca: dry eyes
- Xerostomia: Dry mouth

Often associated with other autoimmune disorders:

- RA
- Raynaud's
- Thyroiditis

Treatment

- Cyclosporine ocular for dry eyes
- Pilocarpine for xerostomia
- Prednisone for severe systemic inflammation
- Avoidance of atropine like drugs

Source: Brandt JE, Priori R, Valesini G, Fairweather D. Sex differences in Sjögren's syndrome: a comprehensive review of immune mechanisms. *Biol Sex Differ*. 2015;6:19. Published 2015 Nov 3. doi:10.1186/s13293-015-0037-7

Summary Slide

- Systemic Lupus Erythematosus, Systemic Sclerosis, and Sjogren’s Syndrome are autoimmune disorders with multi-organ system challenges.

- Understanding of presenting findings and appropriate laboratory tests assist in the diagnosis.

- Treatment is multifactorial – both focusing on symptomatic resolution and limiting disease progression.